PlantPulse

Maintenance management – work instructions: how to do each task, step by step
Paper mill edition

Plant: PM1 · Maintenance department
System: https://app.plantpulse-cmms.com
Version: 12 November 2026 (dashboard colour key and customising, KPI targets, Nelson control charts, condition monitoring alarm levels, company KPIs in the app, root cause analysis, search, management presentation)

Each procedure says **who** does it and **where** in the app, then gives numbered steps. **Tip** boxes save time; **Important** boxes protect people, machines and data.

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1. Getting started

Signing in, installing the app on a phone and finding your way around. For everyone.

1.1 Sign in and change your password

Who: Everyone · **Where:** Any browser

1. Open **https://app.plantpulse-cmms.com** in Chrome (computer or phone). The older address https://84-8-106-22.sslip.io still works.
2. Enter the e-mail and the temporary password the administrator gave you, then press **Sign in**.
3. Change the password at once: tap the **person icon** (top right) › **Change password**.
4. Type the current password, the new password twice, and press **Save**.

Note: A password needs at least 12 characters with three of: lowercase, uppercase, digit, symbol. Never share it: every action is recorded under your name.

1.2 Install the app on a phone

Who: Everyone with a phone · **Where:** Chrome on Android, Safari on iPhone

1. **Android**: open the link in **Chrome**, sign in, then Chrome menu (three dots) › **Install app** or **Add to Home screen**.
2. **iPhone**: open the link in **Safari** (or Chrome), tap the **Share** button › **Add to Home Screen** › **Add**.
3. Open PlantPulse from its **PlantPulse** icon (gear with an orange pulse) on the home screen from now on, and sign in once inside it.

Important: Install from Chrome, not Samsung Internet. Samsung Internet packages the page as an app and Google Play Protect may show a warning. A Google Play version of PlantPulse is in preparation.

Tip: If a new button described in this guide is missing after an update, close the app completely (recent apps list) and open it again.

1.3 Find your way around

Who: Everyone · **Where:** Top bar and menu

1. **Menu** (left, or the three lines on a phone): Dashboard, My work, Scan QR, Work requests, Work orders, Assets, Preventive maintenance, Lubrication, Condition monitoring, Shifts and handover, Spare parts, Shutdowns, KPIs and reports, Administration.
2. **ONLINE** badge: green means connected. Reconnect before saving if it shows offline.
3. **Bell**: your notifications (breakdowns, jobs assigned to you, approvals waiting). Tap one to open the record.
4. **Language button** (next to the bell): English, Arabic (right to left) or Bengali; your choice is remembered on every device.
5. **Magnifier** (top bar, or Ctrl+K on a computer): one search over equipment, work orders, requests, spare parts, plans, measurement points, RCAs and shutdowns. Type a code or a few words in any order (e.g. *crusher motor*).

> **Note:** You only see the menus and buttons your role allows, and if your account has one or more departments (e.g. PM + SP) you only see those areas' equipment, jobs and alerts. If something is missing, ask the administrator; do not use someone else's account.

> **Tip:** Any field where you choose an asset has a small **tree** button: open the levels with the arrows and click a code, or simply type part of the code or name.

1.4 Read the dashboard (colour key)

Who: Everyone · **Where:** Dashboard

1. The **colour key** under the quick buttons applies to the whole dashboard: **red** = act now, **amber** = needs attention or waiting for someone, **green** = all clear or target met, **grey** = information only.
2. **Counters** (open emergencies, overdue, approvals…) turn green with a tick when they are zero. Tap a counter to open exactly the list it counts.
3. **Condition monitoring alarms (open)** counts every open alarm; the line under the number splits it into *alarm limit* and *warning limit*. It is not a plant shutdown.
4. **KPIs: result vs target**: each tile shows the result (last 6 months), the target and **Met** (green), **Close** (amber, missed by up to 10 %) or **Missed** (red). "No target set" = a manager has not set one yet.
5. **Condition monitoring** card: points with an alarm or a Nelson rule signal (R1, R2…, see 3.6 *Read a control chart*). Tap a point to open its chart.

Colour	Meaning	Examples
Red	Act now	Open emergencies, overdue work, overdue lubrication, alarm limit exceeded, KPI target missed
Amber	Needs attention / waiting for someone	New requests, approvals, review, warning limit exceeded, KPI close to target
Green	All clear / target met	Counter at zero, KPI target met, reading normal
Grey	Information	My open work, open work orders, KPI without a target

1.5 Customise your dashboard

Who: Everyone · **Where:** Dashboard › Customise dashboard

1. Press **Customise dashboard** (top right of the dashboard).
2. Under **Sections** tick what you want to see: quick actions, counters, needs attention, work orders per week, KPIs with target and result, condition monitoring signals.
3. Under **Counters** untick the counters you do not need.
4. Under **Calculated KPIs** and **Company KPIs** tick the KPIs to show as tiles.
5. Press **Save**. **Reset to default** brings back the standard layout.

> **Tip:** Your layout is saved on the server for you and this plant, so the phone app and the computer show the same dashboard.

2. Reporting and doing work

From a reported problem to a closed work order: requests, QR, work orders, safety and spare parts.

2.1 Report a problem (work request)

Who: Everyone, including production · **Where:** Work requests › New work request

1. Open **Work requests** › **New work request** (or scan the machine's QR label and press **New work request** on its page).
2. Choose the **asset**: type part of its code or name, or use the tree button.
3. Choose the **Symptom** from the list: what you see or hear (abnormal noise, high vibration, oil leak, tripped…).
4. If you know it, choose the **Suspected failure mode**. The modes of that equipment type are shown first ("For this equipment"). Leave it empty if unsure: the technician confirms it later.
5. The **Title** fills in by itself from your choices; change it only if needed. Write details in **Description**.
6. Tick **Breakdown (machine stopped)** if the machine is stopped; that area's supervisors are notified at once.
7. Press **Submit**. You can follow the request's status in the Work requests list.

> **Tip:** Use a work request instead of a phone call or WhatsApp message, so every breakdown is counted in MTBF and MTTR.

2.2 Scan a QR label

Who: Everyone · **Where:** Scan QR

1. Open **Scan QR** from the menu and press **Start camera**.
2. Point the camera at the label on the machine or lubrication point.
3. The asset's page opens: details, history, spare parts, lubrication, condition monitoring, attachments.
4. To report a fault on it, press **New work request** on that page.

> **Important:** Scan from inside the app. The phone's normal camera app does not open these labels.

> **Tip:** Camera not working? Type the code printed under the QR (cmms:asset:...) in the box on the Scan QR page.

2.3 Accept or reject a work request

Who: Supervisor · **Where:** Work requests

1. Open **Work requests** and open a new request.
2. Check the asset and description; adjust the suggested priority if needed.
3. Press **Accept & create WO** to create the work order, or **Reject** and write the reason.

> **Note:** Breakdowns appear at the top of the Dashboard as open emergencies until they are handled.

2.4 Assign a work order

Who: Supervisor · **Where:** Work orders › open the work order › Details

1. Open **Work orders** (filter by status, type, overdue) and open the job.
2. In the **Details** tab choose the technician(s) under **Assignees** and press **Assign**.
3. The system warns if a technician is on leave or already booked at that time.
4. Several jobs for one technician: tick them in the list and use **Bulk assign**.

> **Note:** Preventive, lubrication and condition-monitoring work orders are created by the system already approved; they only need assigning.

2.5 Carry out a work order

Who: Technician · **Where:** My work › open the job

1. Open **My work**. Jobs are sorted by due date; a warning mark means LOTO or a permit is needed.
2. If LOTO is needed, record every isolation point (see the next procedure) and wait for the supervisor's sign-off.
3. Press **Move to: In progress**.
4. In the **Checklist** tab mark each task OK or Not OK and enter readings; out-of-range readings are flagged.
5. Record findings and downtime, and choose **Symptom**, **Problem** (the failure mode; this equipment's modes are listed first), **Cause** and **Remedy**. Add photos in **Attachments** (the camera opens on a phone).
6. **Labor** tab: **Book labor** for your own hours.
7. **Materials** tab: record every spare part you fitted (see "Record spare parts used").
8. Remove your locks, then press **Move to: Awaiting review**.

> **Important:** Book your hours on every job. Without labour hours, Planned Maintenance %, Emergency % and labour cost stay "Not available".

2.6 Record LOTO and sign off the permit

Who: Technician (isolation), supervisor (sign-off) · **Where:** Work order › Checklist › LOTO isolation register

1. Technician: for each isolation point enter the point (e.g. MCC-3 breaker Q12), the energy type and the lock / tag number, then press **Apply lock**.
2. After testing, tick **Zero energy verified** (**Verify**) on each point.
3. Supervisor: check all points, enter the permit number, permit types (cold work, hot work, electrical, confined space, work at height...) and **Permit valid until**.
4. Supervisor: press **Sign off LOTO / permit**. Only now can the job move to In progress.
5. When the work is finished, the technician presses **Remove lock** on each point.

Important: The system refuses: starting work before sign-off, signing off an expired permit, and completing the job while a lock is still applied. It supports the physical lock-and-tag procedure; it does not replace it.

2.7 Record spare parts used

Who: Technician or supervisor · **Where:** Work order › Materials › Spare parts used

1. Open the work order's **Materials** tab.
2. In **Spare part**, type the part code or part of the description and pick it.
3. Not in the list? Type a description in **Item not in the list** instead.
4. Enter the **Quantity** (unit cost is optional; blank uses the book cost) and press **Add part used**.

Note: This records which part went on which job and machine, and adds its cost to the work order. Store stock is not changed, because the store system is not connected.

Tip: Spare parts › **Parts used** shows all parts by machine (including everything below it), by part and by period, with Excel export.

2.8 Review and close a work order

Who: Supervisor · **Where:** Work orders › status Awaiting review

1. Open the Dashboard counter **Awaiting review**, or filter Work orders by that status.
2. Check the checklist, readings, hours, parts, photos and findings.
3. Record a test result if the job needs one (**Record result**).
4. Use **Move to** to complete and close the job, or send it back if something is missing.

Note: Closing is refused until required checklist items, test results and failure codes are complete (root cause too for emergencies). The message lists what is missing. Each work order can be printed as a PDF report.

3. Planned work

Preventive maintenance, lubrication and condition monitoring create their work orders automatically.

3.1 Follow the PM schedule

Who: Planner, supervisor · **Where:** Preventive maintenance

1. **PM plans** tab: all plans with last done and next due; use the search box for a code or name; use the page numbers at the bottom for more.
2. **Schedule** tab: coming due dates with plan, machine and a link to the work order once it is created.
3. Work orders are created up to 7 days before the due date by the scheduler, which runs every 15 minutes. **Run scheduler now** forces a run.
4. **Job plans** tab: the task list, hours and safety needs behind each plan.

> **Tip:** A monthly mechanical round on one machine is one work order with all its tasks in the checklist.

3.2 Move or cancel a PM

Who: Anyone with PM access (request), supervisor (approval) · **Where:** Preventive maintenance › Schedule

1. Find the due date in the **Schedule** tab.
2. Press **Request deferral** (choose the new due date) or **Request cancellation**, and write the reason.
3. A supervisor opens the **Approvals** tab and approves or rejects it. You cannot approve your own request.

> **Important:** Never just leave a PM open. An unexplained overdue PM lowers PM compliance.

3.3 Import a plan from Excel (PM, lubrication or condition monitoring)

Who: Planner (PM, lubrication), reliability engineer (condition monitoring) · **Where:** Import plan button on the Preventive maintenance, Lubrication or Condition monitoring page

1. Press **Import plan** and download the template.
2. Fill one row per task (PM), per lubrication point, or per measurement point. Leave **start_date** blank to spread the first due dates over the first cycle.
3. Choose the file and press **Preview**. It shows how many rows are valid and lists every error.
4. Fix any errors (download the error report if needed), choose the file again and **Preview** again.
5. When there are 0 errors, press **Import**.

> **Note:** Import is all or nothing: if one row has an error, nothing is saved. A plan or point code that already exists is reported as an error, so importing the same file twice never doubles the data.

3.4 Do a lubrication route

Who: Lubricator / technician · **Where:** My work › lubrication work order

1. Each route gives one work order with the points due in the next 2 days. Open it from **My work**.
2. In the **Checklist** tab, for each point: enter the actual quantity, the lubricant used and a condition note, then press **Record** (or **Skip** with a reason).
3. Complete the work order when all points are recorded.

Important: Use only the lubricant shown on the point, or an alternative approved in **Lubrication › Approve alternative**. Any other lubricant is refused.

Tip: Replace the general names “General oil / grease (specify product)” with the real products in the **Lubricants** tab.

3.5 Enter a condition reading and handle an alert

Who: Technician (reading), reliability engineer (alert) · **Where:** Condition monitoring

1. Open **Condition monitoring** › **Measurement points**. Critical and High points are at the top (red / orange badge); tick **Critical & high only** to see just those.
2. Find a point with **Search** (point, equipment code or name) or **Asset and everything below it**, then click it. On a phone it opens full screen.
3. Enter the value in **Record reading** and press **Save**. The level becomes **Normal** (green), **Warning limit** (amber) or **Alarm limit** (red) against the limits set for the point. Close the panel with the **X** at its top right.
4. A reading outside a limit opens an alarm in the **Alerts** tab, sent to that area’s engineers. Filter the tab with **All open** / **Alarm limit exceeded** / **Warning limit exceeded**.
5. The engineer presses **Acknowledge**, then **Create work order** if action is needed, and finally **Resolve** with the resolution.

Note: Status comes from clear limit rules; the system does not predict failures. A point not read for too long becomes “unknown”. The screen list shows the first 2,000 points; use search for the others.

3.6 Read a control chart (Nelson rules)

Who: Reliability engineer, engineer, supervisor, manager · **Where:** Condition monitoring › click a point; KPIs and reports › click a KPI or a company KPI

1. The chart shows the readings (or monthly results) in time order. Dots are coloured by level: green normal, amber warning limit, red alarm limit (for KPIs: target met / close / missed).
2. **CL** (thick green line) is the average; **UCL** and **LCL** (dark red lines, with their values) are 3 standard deviations (σ) above and below. Zones: **C** within 1σ, **B** 1–2σ, **A** 2–3σ. Dashed orange and red lines are the point's warning and alarm limits.
3. A point inside a broken rule is an **open red circle**; the **red number** next to a point is the rule whose pattern starts there.
4. The table under the chart lists the 8 rules with their probable cause; broken rules are red with the dates (first → last).
5. In the point list, the **Nelson** column shows the rules broken now (the last 30 readings, pattern ending in the last 3).

Rule	Pattern	Probable cause
1	1 point more than 3σ from CL	New person, wrong setup
2	7 points in a row on the same side of CL	Setup change, process change
3	7 points in a row all increasing or all decreasing	Trend, tool wear
4	14 points in a row alternating up and down	Over-control, tampering
5	2 out of 3 points more than 2σ from CL (same side)	New person, wrong setup
6	4 out of 5 points more than 1σ from CL (same side)	Small shift (similar to rules 1 and 5)
7	14 points in a row within 1σ of CL (either side)	Process change
8	8 points in a row more than 1σ from CL (either side)	Process change

Note: A rule signal means the process changed, even when no limit is passed: look for the cause before it becomes an alarm. At least 3 values are needed; σ comes from the average moving range of the values on the chart (last 100 readings).

3.7 Set the priority of measurement points

Who: Planner, engineer, reliability engineer, manager, administrator · **Where:** Condition monitoring

1. One point: click it and choose **Critical**, **High** or **Normal** next to **Priority**.
2. Many points: press **Priorities (Excel)** › **Download points with priorities**.
3. In the yellow **priority** column choose Critical, High or Normal (filter and sort freely; only that column is read).
4. Upload the file, press **Preview** (shows how many change), then **Apply priorities**.

Tip: Example: all bearing points of the 73 dryer cylinders were set to Critical in one upload.

3.8 Add a measurement point (and missing equipment)

Who: Reliability engineer · **Where:** Condition monitoring › New measurement point

1. Choose the **Asset** (type or use the tree).
2. Equipment not in the list? Press **+ Equipment not in the list? Add it** (or **+ Add equipment under the selected asset**), fill code and name, **Create**.
3. Choose the **Technique** and **Unit**; enter warning and alarm limits.
4. Leave **Code** and **Name** empty: they are made from the asset (e.g. PM-DRYER-04-CM27). Press **Create**.

3.9 Import past readings (history)

Who: Reliability engineer · **Where:** Condition monitoring › Import readings history

1. Press **Import readings history** › **Download template**. Its **points** sheet lists every point code.
2. Fill one row per reading: **point_code**, **date**, **time** (optional, blank = 08:00), **value**, **note**.
3. Choose the file, press **Preview** (readings, points, period, how many critical / warning), then **Import**.

> **Note:** Old readings appear in the trend and get a status from today's limits, but raise no alerts. A point's last value changes only if the imported reading is newer. Importing the same file twice adds nothing.

4. Assets and labels

Finding machines, their history, equipment types and printing QR labels.

4.1 Find an asset and its history

Who: Everyone · **Where:** Assets

1. Open **Assets**. **Hierarchy** shows the tree (area › line › unit › equipment); **List** lets you search by code or name.
2. Open the asset. Tabs: **Details**, **History** (every job, including migrated 2026 jobs marked MIG), **BOM / spare parts**, **Meters / Condition monitoring**, **Lubrication**, **Attachments**.
3. The **Reliability** card shows MTBF, MTTR, availability and breakdown downtime for 12 months.

4.2 Set equipment types (which failure modes apply)

Who: Planner, engineer · **Where:** Assets › Equipment types; asset page › Edit

1. Press **Equipment types** › **Preview**: the system suggests a type from each name (pump, motor, gearbox, roll, dryer cylinder, crane…) for assets without one.
2. Check the samples, then press **Apply to … assets**.
3. For an asset it could not recognise: open it › **Edit** › **Equipment type**. Parts below an asset use its type.

> **Note:** The type decides which failure modes are offered first on requests and work orders for that machine.

4.3 Print QR labels (A4, 24 labels per sheet)

Who: Planner, supervisor · **Where:** Assets › Print QR labels (also on each asset page and on Lubrication)

1. Open **Assets** and press **Print QR labels**.
2. In **Asset and everything below it** choose an area to start with, e.g. Press.
3. Leave **Level** empty for all levels, or choose Production line and Unit for the main machines only.
4. If the first sheet is partly used, enter how many labels are already used in **Labels already used on the first sheet**.
5. Press **Download PDF**.
6. First time: print one page on plain paper at **Actual size / 100%** (not “Fit to page”), lay it on the label sheet against the light and check the positions.
7. If it matches, print on the label sheets. If not, send the size written on the label box to the administrator.
8. Stick a label on the machine and test it with **Scan QR**.

Note: Sheet: 3 × 8 labels of 70 × 37 mm, no margin. Each label shows the QR, the code, the name and the location. At most 2,400 labels (100 sheets) per PDF; narrow the selection if asked.

Tip: On the Lubrication page, labels can be printed per route; each shows the lubricant, quantity and frequency.

5. Shifts and roster

Shift types and the monthly roster from Excel.

5.1 Set up shift types

Who: Manager, planner · **Where:** Shifts and handover › Shift types

1. Press **New shift type** and add each code the roster uses, e.g. **D** Day 06:00–18:00 12 h, **N** Night 18:00–06:00 12 h, **G** General 07:00–17:00 10 h.
2. Use your real times and hours; the hours count as available manpower.

5.2 Import the roster from Excel

Who: Manager, planner · **Where:** Shifts and handover › Import roster (Excel)

1. Set **From** and **To** (up to 62 days, e.g. a whole month) and press **Download roster template**.
2. Press **Enable Editing** in Excel. Each day cell has a drop-down: D, N, G (your shift types), OFF, L (leave), A (absence). Fridays and Saturdays are shaded but can still be duty days. Blank = no change.
3. Save, choose the file, press **Preview** (people, entries, dates, unknown names), then **Import**.

Note: If the drop-down shows only OFF, L and A, the shift types are not set up yet: add them, then download the template again. Your own grid also works: a row starting with “Name”, then the days (day numbers or dates), several blocks allowed.

Tip: Names not in the system can be created as technicians by the administrator (tick “Create them as technicians”); they get a login after the administrator sets their e-mail and resets the password.

6. KPIs, targets and company KPIs

Calculated KPIs and their targets, failure modes, the plant's own monthly KPI sheet and the management presentation.

6.1 View KPIs for a period and department

Who: Manager, engineer, supervisor · **Where:** KPIs and reports › KPIs and reports

1. Set **From** and **To**. The page opens on the current month; set 1 January for the whole year.
2. Choose a **Department** or leave it empty for the whole plant.
3. Each card is coloured against its target (colour key above the cards): green met, amber close (missed by up to 10 %), red missed, white no target.
4. Tap a KPI to see its formula, inclusion rules, the 12-month control chart with the target, and the work orders behind it.
5. Use **Export XLSX** or **Export PDF** for reports.

> **Note:** "Not available" means the data is missing, and the reason is shown. Values are never estimated. Maintenance cost today is spare parts only.

6.2 Set or change a KPI target

Who: Maintenance manager (KPI management) · **Where:** KPIs and reports › KPIs and reports

1. For a **whole-plant** target leave **Department** empty; for one department choose it first at the top of the page.
2. On the KPI card press **Set target** (or **Change** when a target exists).
3. Type the **Target**. **Better when** is preset (higher for PM compliance, MTBF, availability; lower for MTTR, backlog, costs); change it only if needed.
4. Press **Save**. The card, the dashboard tiles and the management presentation use the target at once. **Remove target** deletes it.

> **Note:** Which target is used: the department's own target, else the plant target, else the target of a company KPI linked to this KPI.

6.3 See the top failure modes (Pareto)

Who: Manager, reliability engineer, supervisor · **Where:** KPIs and reports › Failure modes

1. Set the period at the top of the KPIs page and open the **Failure modes** tab.
2. Choose **By failure mode** (adds up the same mode on all equipment types), **By code** or **By symptom**.
3. Optionally choose an asset to see only it and everything below it.
4. The chart shows the top 10; the table gives work orders, downtime hours, share and cumulative %.

> **Note:** Only work orders with a failure code are counted. The orange note shows how many closed breakdowns have no code: remind technicians to choose the Problem before completing.

6.4 Enter production data and budget

Who: Manager or planner · **Where:** KPIs and reports › Production data / Budget

1. Production: choose the date and department, enter **Tons produced**, **Good tons**, **Planned hours**, **Run hours** and **Ideal rate**, and save.
2. Budget: choose department, year and month, enter the amount, and save.

> **Tip:** These unlock OEE, maintenance cost per ton and budget versus actual.

6.5 Add a company KPI in the app (no Excel)

Who: Manager · **Where:** KPIs and reports › Company KPIs

1. Open the **Company KPIs** tab and press **Add KPI**.
2. Type the **KPI name**; choose a **Department** (blank = whole plant), the **Unit** and **Better when** higher or lower.
3. Optional: **Link to a CMMS KPI** (e.g. MTBF) to show its target next to the calculated value.
4. Press **Save**, then **Enter month** to type the first results. **Edit** on a row changes the KPI; **Delete** removes it with its values.

6.6 Enter a month of company KPIs (in the app)

Who: Manager · **Where:** KPIs and reports › Company KPIs

1. Open the **Company KPIs** tab and choose the **Year**.
2. Press **Enter month** and choose the month (e.g. Oct 2026).
3. Type the **Actual** for each KPI. The **Target** is filled in from the previous month; change it only if the target changed.
4. Press **Save**. Each month is coloured: green met, amber close (missed by up to 10 %), red missed. Click the KPI name for its control chart.

> **Tip:** Open the same month again to correct a value. Clearing a box deletes that value.

6.7 Update company KPIs from Excel

Who: Manager · **Where:** KPIs and reports › Company KPIs

1. Choose the **Year** and press **Download template**; it comes back filled with the saved values.
2. Fill or change the **Actual** and **Target** rows under the month columns (YYYY-MM).
3. Choose the file under **Upload filled template**, press **Preview**, then **Import**.

> **Note:** A blank cell keeps the saved value; type x to delete a value. A new KPI name in the file adds a new KPI. Columns: department code (blank = whole plant), unit, higher_is_better (yes / no), cmms_kpi (optional link to a calculated KPI such as mtbf).

6.8 Make the management presentation

Who: Maintenance manager, planner · **Where:** KPIs and reports › Management presentation

1. Set **From** and **To** (a whole month, quarter or year compares with the same length before) and, optionally, a **Department**.
2. Press **Management presentation**. A PowerPoint file in your interface language (English, Arabic or Bengali) is downloaded.
3. Slides: executive summary, KPI scorecard (this period, previous period, target, status), 12-month trends, work orders, top 5 bad actors, cost against budget, emergency breakdowns, RCAs, lubrication and condition monitoring (alarm levels and Nelson signals), plan for the next period.
4. Status on the scorecard: **On target** (green), **Close to target** (amber, missed by up to 10 %), **Off target** (red); without a target, better / worse than the previous period.

> **Tip:** Charts are real PowerPoint charts: add comments or change colours before the meeting. Set the KPI targets first so the scorecard shows them.

7. Root cause analysis

Major breakdowns are investigated with 5 Whys and a fishbone, and the actions are followed until they prove effective.

7.1 Decide whether an RCA is needed

Who: Maintenance manager (decision); engineer or supervisor (request) · **Where:** Root cause analysis; or a work order › Request RCA

1. An RCA is proposed automatically when a completed breakdown matches the **RCA rules** (long downtime for the asset's criticality, the same failure mode repeating, a safety or environment event, high cost or production loss). Engineers and supervisors can also press **Request RCA** on a work order.
2. The manager is notified. Open **Root cause analysis**, open the RCA **Waiting for decision**.
3. Press **Start RCA**: choose the leader (an engineer), the team members and the due date (default 7 days); or press **Not required** with a reason.

7.2 Carry out the RCA (standard order)

Who: RCA leader and team · **Where:** Root cause analysis › open the RCA – the tabs are numbered in this order

1. **1. Problem (5W1H)**: What, Where, When, Who, Which / how much, How – check the text pre-filled from the work orders, write the problem statement and add the timeline. ("Why" is not asked here: it is step 4.)
2. **2. Evidence & data**: photos, readings, the failed part, reports. Facts first, before guessing causes.
3. **3. Fishbone**: list every possible cause under Man, Machine, Method, Material, Measurement, Environment; check each with the evidence and mark it verified or ruled out.
4. **4. 5 Whys**: press **Ask why?** on a verified cause (or start from the problem) and keep asking "why?". Five is a guide, not a limit: add Why 6, 7… until the answer is a cause you can act on. Add another answer for a second branch.
5. **Root cause = the last why** of each chain: tick **Root cause** there (it is only possible on the last why). Asking why again moves the mark down.
6. **5. Root causes**: press **Use** on each marked last why and set its type (physical, human, system).

7 **6. Action plan**: containment, corrective and preventive actions with owner and due date, each linked to a root cause. Press **Save**, then **Submit for approval**.

> **Note:** Submitting is refused until there is a problem description, a 5 Whys chain with its root cause marked on the last why, at least one root cause and at least one action. Step 7, the effectiveness check, follows the actions.

7.3 Approve, act and check the effect

Who: Maintenance manager (approval); action owners · **Where:** Root cause analysis › open the RCA

1 The manager **Approves** the RCA or returns it to the team with a note.
2 Owners finish their actions with the result; **Create work order** raises a work order for an action, and completing that work order finishes the action.
3 When the last action is done, the **effectiveness check** runs (default 90 days). If the same failure comes back on the asset, the RCA reopens by itself.
4 At the end the manager closes the RCA as effective. **Print report** gives a one-page A4 report.

8. Roles and approvals

What each role can do and who approves each step. The chart and table follow the procedures of this part.

8.1 Find who approves a work order

Who: Everyone · **Where:** Work order › Move to

1 Open the work order: the **Move to** buttons show only the steps your role may do.
2 Breakdown requests: a supervisor or planner of that area accepts them; the work order is created at once.
3 Corrective work: **Send for approval**, then a supervisor or the manager **approves** (or returns it with a question).
4 The planner plans and schedules, the technician does the work and submits it for review, the supervisor or engineer reviews, the supervisor or manager closes.

> **Note:** Emergency breakdowns skip approval: the technician starts at once and the area's supervisors and manager are alerted by app and e-mail. PM and lubrication work orders are created already approved.

> **Tip:** Administrators can change which role may do each step in Administration › Workflow. The safety and completion checks always apply.

Role	Can do / approve
System Administrator	Everything; the only role that manages users, departments, plant settings and workflow
Maintenance Manager	Every maintenance step: approve, plan, close, emergency closure, PM deferrals, KPIs and budgets
Maintenance Planner	Triage requests, create / plan / schedule / assign work orders, PM and lubrication plans, imports. Does not approve, execute or close

Role	Can do / approve
Mechanical / Electrical Supervisor	Triage, approve, plan, assign, execute, review, LOTO / permit sign-off, close (incl. emergencies), cancel, PM deferrals, handovers
Reliability Engineer	Create work orders, review completed work, condition monitoring points and alerts, PM and lubrication plans, lubricant alternatives
Engineer	Create work orders, do and review work, readings, point priority, edit assets
Technician	Work requests, do the work, lubrication tasks, readings, handovers (no approvals, no KPIs)
Storekeeper	Spare parts master and stock, reservations, purchase references
Production User	Work requests only; sees assets, work orders, condition monitoring, KPIs
Management Viewer	Sees everything read-only, exports reports

PlantPulse – Work request & work order approval flow

Each step shows the status and the roles allowed to move it on. Users limited to departments act only on their own departments' equipment. System Administrator can do every step.

1. Work request submitted
Equipment, standard symptom, photo, breakdown yes/no.
Technician · Production User · Engineer · Reliability Eng. · Supervisors · Planner · Manager

↓ Triage: accept → work order created | reject (reason required)

2. Triage
Supervisors · Planner · Manager
Work orders can also be raised directly by Planner, Supervisors, Engineer, Reliability Eng., Manager.

↓ Send for approval: Planner, Supervisors, Engineer, Reliability Eng., Manager

3. Awaiting approval
Approve, or return to Triage for more information.
Mech. Supervisor · Elec. Supervisor · Manager

↓

4. Approved → Planned → Scheduled
Plan labour & parts, assign technicians; Scheduled needs a planned start date.
Planner · Supervisors · Manager

↓ If LOTO / permit required: safety sign-off first — Supervisors, Manager

5. In progress ⇄ On hold
Labour, checklist, parts, findings, failure codes, photos.
Technician · Engineer · Supervisors · Manager

↓ Submit for review – the assigned supervisor is notified

6. Awaiting review (inspection / test)
Accept as Completed, or send back for rework (→ In progress).
Supervisors · Engineer · Reliability Eng. · Manager

↓

7. Completed
System checks before completing & closing (see right).

↓ Technical closure

8. Closed
Mech. Supervisor · Elec. Supervisor · Manager

⚡ Emergency breakdown
Breakdown request accepted, or emergency WO raised directly.
Skips approval: technician starts at once (Triage → In progress).
Instant alert (app + e-mail) to supervisors & manager of the area.
Technician · Engineer · Supervisors · Manager

↓ joins at "In progress"

Emergency closure
Root cause required. Only:
Mech. Supervisor · Elec. Supervisor · Manager

PM & lubrication WOs
Generated automatically by the scheduler (lead days before due) – they start already **Approved**.

Cancel (any open step)
Reason required.
Planner · Supervisors · Manager
Cancelling a PM work order also needs PM-deferral approval (Supervisors, Manager).

Who gets notified
- Awaiting approval → approvers of that department
- Submitted for review → assigned supervisor (or reviewers of the area)
- Emergency → supervisors & manager, also by e-mail

Completion checks (cannot be skipped)
- All LOTO locks removed
- LOTO / permit signed off; permit reference entered
- Test / inspection passed (when required)
- All required checklist items done
- Problem, cause & remedy codes (corrective & emergency)
- Lubrication tasks of the WO recorded
- Root cause for emergency breakdowns

Configurable
Administrators can switch individual steps on/off and change which permission each step needs (Administration → Workflow). The checks above always apply.

Other approvals

PM deferral / cancel	**Request to defer or cancel** a due PM (new date + reason) — Anyone who sees PM	→ **Approve / reject** — Supervisors · Manager	→ **Due date moved** or occurrence cancelled; history kept for PM compliance
Condition monitoring alert	**Reading recorded** above warning / alarm limit — Technician · Engineer · Reliability Eng. · Supervisors · Manager	→ **Alert** to the area (Critical can raise a WO automatically) Acknowledge / resolve: Reliability Eng. · Supervisors · Manager	→ **Work order** raised from the alert → normal WO flow
Lubricant alternative	**Alternative lubricant** proposed for a point	→ **Approve** (compatibility verified) — Reliability Eng. · Manager	→ **Allowed** on lubrication tasks
Shift handover	**Write handover** (open breakdowns listed automatically) — Technician · Engineer · Supervisors	→ **Confirm** by incoming shift — Supervisors · Manager	→ **Handover closed**
Users & access	**New user / role / departments**	→ **Only** System Administrator (only a System Admin can make another System Admin)	→ **Recorded** in the audit log

9. Administration

For the PlantPulse administrator only.

9.1 Add, edit, reset or remove a user

Who: Administrator · **Where:** Administration › Users

1. **New user**: name, e-mail, role, optional departments, trade and hourly rate (used for labour cost), temporary password. Save.
2. **Departments** column in the Users list: pick one or more (PM, SP, FH, UTIL, WWTP, EAI, WS, HVAC, MOB), e.g. **PM + SP** for someone who covers two areas; the first is the main one. The person then sees only those areas (lists, alerts, KPIs, notifications). The **X** at the right of the field clears it = All departments. Leave managers, planners and reliability engineers on All.
3. **Edit user**: change name, role, departments, trade or rate.
4. **Reset password**: give the new temporary password to the user; they change it after signing in.
5. **Disable** stops the user at once; **Remove from plant** removes them (their name stays on past records; the e-mail can be added again later).

9.2 Import many users from Excel

Who: Administrator · **Where:** Administration › Users › Import users (Excel)

1. Press **Import users (Excel)** › **Download current users**: the file lists everyone already in the plant.
2. Change any row (role, department, trade, e-mail) or add new people below. Several departments: write them like **PM,SP**. Leave **old_email** blank on new rows; write **yes** in **reset_password** to give someone a new temporary password.
3. Choose the file and press **Preview**: new, changed, unchanged and every error by row.
4. Press **Import**. The temporary passwords of new people are shown once: press **Save passwords (CSV)** and give each person their own, privately.

> **Important:** Store the passwords file safely and delete it once handed out. Nobody is removed by an import.

9.3 Try changes on the test copy first

Who: Administrator · **Where:** Server terminal (PowerShell or Cloud Shell)

1. Connect to the server and go to **/opt/cmms**.
2. **sudo bash scripts/test-copy.sh start** creates a copy of today's real data at **https://test.84-8-106-22.sslip.io** (red TEST label, no e-mails sent).
3. Try imports, new settings or training there.
4. **refresh** copies the real data again; **stop** pauses it; **remove** deletes it.

> **Note:** Nothing done on the test copy reaches the real system.

9.4 Departments and failure codes

Who: Administrator · **Where:** Administration › Departments / Failure codes

1. **Departments**: enter code, name and type, press **Add** (e.g. WS Central Workshop, HVAC Air Conditioning, MOB Mobile Equipment, type UTIL).
2. **Failure codes**: press **Import failure codes (Excel)**; download the current list, add or change rows (kind: symptom / problem / cause / remedy, code, description, equipment_type, category), upload, **Preview**, **Import**.
3. Use **Search** to find a code (e.g. PMP, ROL, ACU, MOB, bearing).

Note: Existing codes are updated, codes not in the file stay. Loaded so far: failure modes for 12 general and 12 paper-machine equipment types, air conditioners, forklifts and machine tools, plus standard symptoms, causes and remedies.

9.5 Update the server to a new version

Who: Administrator · **Where:** PowerShell on a laptop, or Oracle Cloud Shell

1. Open **PowerShell** on your laptop, or sign in to **cloud.oracle.com** and open **Cloud Shell** (the >_ icon at the top).
2. Paste the update command below and press Enter.
3. Wait until **==> Done** appears (a few minutes). Do not switch apps or lock the phone meanwhile.
4. If the screen stops or shows “Connection closed” before Done, run the same command again; it is safe.
5. Ask users to refresh the page or reopen the app.

```
ssh ubuntu@84.8.106.22 'cd /opt/cmms && sudo ENV_FILE=deploy/.env scripts/backup.sh
/opt/cmms-backups && sudo BRANCH=ccr-09be8126-0y17zi bash deploy/install-server.sh'
```

Note: The command first takes a backup, then fetches the version named after BRANCH (the administrator is told which one), applies database changes and restarts. The old version keeps running until the new one is ready.

Tip: Check afterwards: **sudo docker compose -f deploy/docker-compose.yml --env-file deploy/.env ps** shows api, web, worker and db as Up (healthy).

9.6 Check the backups

Who: Administrator · **Where:** Oracle console › Storage › Buckets

1. Every night at 02:15 the server backs up the database and attachments and uploads a copy to the bucket.
2. Open **Storage › Buckets** › the CMMS bucket › **Objects**; each night adds a dated folder with 4 files.
3. Once a month, open the latest folder and **Download** the cmms_....dump file to a company drive.

Important: Never share the bucket upload link (pre-authenticated request). Anyone holding it can write to the bucket.

Note: Free storage is 10 GB. Check the bucket’s usage every few months and delete old folders if it approaches 8 GB.

10. Troubleshooting

Quick answers to the most common problems.

Problem	What to do
A button from this guide is missing	Refresh the page; on a phone close the app from the recent apps list and open it again. If still missing, your role may not include it.
Play Protect warning when installing	Install from Chrome (menu › Add to Home screen), not from Samsung Internet.
The phone camera does not open a QR label	Use Scan QR inside the app, or type the code printed under the QR.
A KPI shows “Not available”	Read the reason on the card: usually missing labour hours, production data, budget or rosters.
An import is refused	Preview lists every error by row; fix them, choose the file again and preview again. Nothing is saved until all rows are valid.
I cannot start a job	LOTO and permit must be signed off first; the message names what is missing.
I cannot close a job	The message lists what is missing: checklist items, test result, failure code, root cause or a lock still applied.
Cloud Shell stopped updating during a server update	Wait 3 minutes, reload the page, open Cloud Shell and run the update command again.
Forgot password	Ask the administrator to use Reset password in Administration › Users.
I only see my own area	Your account has departments. Ask the administrator to add the other areas you cover, or to clear the field (All departments) if you cover the whole plant.
Roster drop-down shows only OFF, L, A	Add the shift types (D, N, G…) in Shifts and handover › Shift types, then download the template again.
A point or asset is not in the list	Lists show the first 2,000 items: use Search or the asset tree. Excel downloads always include everything.
iPhone: no Install button	Use the Share button › Add to Home Screen (in Safari, or Chrome’s share button).
A KPI tile says “No target set”	A manager sets it: KPIs and reports › Set target on the KPI card (whole plant, or a department chosen at the top).
The control chart says at least 3 values are needed	Enter more readings (or more months for a KPI); the rules need history to judge the process.

Problem	What to do
The dashboard looks different on my phone	Customise dashboard saves your layout on the server: open the dashboard again, or press Reset to default.